Log 708 - Chapter 4 Solutions

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4.1

a. We can use the test statistic

$$T = \frac{\bar{x} - 290}{S/\sqrt{n}}$$

and use $N(0,1)$ as null distribution. We calculate the observed value either with a calculator or as here, using R

```
t_obs <- (319 - 290)/(100/sqrt(400))
t_obs
```

```
## [1] 5.8
```

The observed value is 5.8. Now, since the test is two-sided, we get the P-value as

$$P = P[T \leq -5.8 \text{ or } T \geq 5.8] = 2 \cdot P[T \leq -5.8]$$

```
pval <- 2*pnorm(-5.8)
pval
```

```
## [1] 6.631492e-09
```

As expected from the observed T value, the P-value is practically 0. We reject H_0 at the given level.

b. For this we use the binomial test. We first note that

$$\hat{p} = \frac{56}{400} = 0.14$$

Here the test statistic and observed value becomes

$$Z_{OBS} = \frac{\hat{p} - 0.20}{\sqrt{\frac{0.20(1-0.20)}{n}}} = \frac{0.14 - 0.20}{\sqrt{\frac{0.20(1-0.20)}{400}}} = -3.0$$

The P-value is now $P = 2 \cdot P[Z \leq -3.0] = 0.003$, so again we reject the null hypothesis. The proportion of cash paying customers is likely to be quite a bit lower than the suggested 0.20.

c. Ok, by assumption in this part, we have 20% paying with cash and they pay on average an estimated 319 NOK. On the other hand, 10% make cash withdrawals at average 400 NOK. I.e. on average, for each withdrawal (-400) there are two cash payments at $2 \cdot 319 = 638$ NOK, so the balance should be OK. (Even with the estimated 14% paying cash, we are OK, because each -400 withdrawal is countered by $1.4 \cdot 319 = 447$ NOK in cash payment.)

4.2

We start by reading the data and look at top rows

```
tripdata <- read.csv("M:/Undervisning/Undervisningh25/Log708/Data/Trip_durations.csv")

head(tripdata)
```

```
##   Duration Distance
## 1     12.5     2.74
## 2     33.5    13.60
## 3     33.1    10.99
## 4     37.0    10.31
## 5     18.3     4.93
## 6     29.3     8.15
```

a. We calculate n, $\bar{x}$, S_x as follows. (Note, we can name the code chunks which makes finding errors a little easier sometimes)

```
n <- nrow(tripdata)
x_bar <- mean(tripdata$Duration)
s_x <- sd(tripdata$Duration)
#print the three numbers as a vector (just to save space in output)
c(n, x_bar, s_x)
```

```
## [1] 200.000000  24.310500   6.835471
```

So $n = 200, \bar{x} = 24.31, S_x = 6.84$.

b. The test statistic should be

$$T = \frac{\bar{x} - 24}{S/\sqrt{n}}$$

and the null distribution (the distribution assuming H_0 is true) is a t-distribution with $df = n - 1 = 199$. We can approximate this very well with the $N(0, 1)$ distribution. We can calculate the T_{obs} value:

```
t_obs <- (x_bar - 24)/ (s_x / sqrt(n))
t_obs
```

```
## [1] 0.6424039
```

We get $T_{obs} = 0.64$. The test is two-sided, so we get the P-value as follows.

$$P = P[T \leq -0.64 \text{ or } T \geq 0.64] = 2 * P[T \leq -0.64].$$

Using R and the $N(0, 1)$ distribution, we get

```
p_value <- 2*pnorm(-0.64)
p_value
```

```
## [1] 0.5221726
```

The P-value is about 0.52 and with significance level $\alpha = 0.05$ we can not reject the H_0.

c. Now we can run with the `t.test` function from R.

```
durtest1 <- t.test(tripdata$Duration, mu = 24, alternative = "two.sided")
durtest1$statistic
```

```
##         t
## 0.6424039
```

```
durtest1$p.value
```

```
## [1] 0.5213503
```

We see that we get almost identical values.

d. This means we simply have to change the constant in the test to 23 and repeat. The test statistic is

```
t_obs <- (x_bar - 23)/ (s_x / sqrt(n))
t_obs
```

```
## [1] 2.711338
```

Using R and the $N(0, 1)$ distribution, we get a new P-value:

```
p_value <- 2*pnorm(-2.71)
p_value
```

```
## [1] 0.006728321
```

And in this case we clearly reject H_0. We can confirm this by testing directly with R:

```
durtest2 <- t.test(tripdata$Duration, mu = 23, alternative = "two.sided")
durtest2$statistic
```

```
##        t
## 2.711338
```

```
durtest2$p.value
```

```
## [1] 0.007287267
```

Again, the results are very close, and the same conclusion applies.

e. Since we are to seek evidence for $\mu_s > 10$, this must be the alternative hypothesis H_1, and then we can have $H_0 : \mu_s = 10$ as the null hypothesis. This is then a one-sample t test with a one-sided alternative, that the mean is *greater* than 10, so in R we can do

```
disttest <- t.test(tripdata$Distance, mu = 10, alternative = "greater")
#now we can chech the whole output.
disttest
```

```
##
##  One Sample t-test
##
## data:  tripdata$Distance
## t = 1.9522, df = 199, p-value = 0.02616
## alternative hypothesis: true mean is greater than 10
## 95 percent confidence interval:
##  10.06158      Inf
## sample estimates:
## mean of x
##  10.40115
```

We get $T_{obs} = 1.95$ and a resulting P-value at 0.026. This is below 0.05, so we reject H_0.

f. One (out of many) ways to get a 95 confidence interval for a mean in R is to run the t.test with a twosided alternative. Since this is the "default" setting for `t.test`, and also the `mu` value is irrelevant for the confidence intervals, we can simply do

```
dur_test <- t.test(tripdata$Duration)
dist_test <- t.test(tripdata$Distance)

dur_test$conf.int
```

```
## [1] 23.35737 25.26363
## attr(,"conf.level")
## [1] 0.95
```

```
dist_test$conf.int
```

```
## [1]  9.995949 10.806351
## attr(,"conf.level")
## [1] 0.95
```

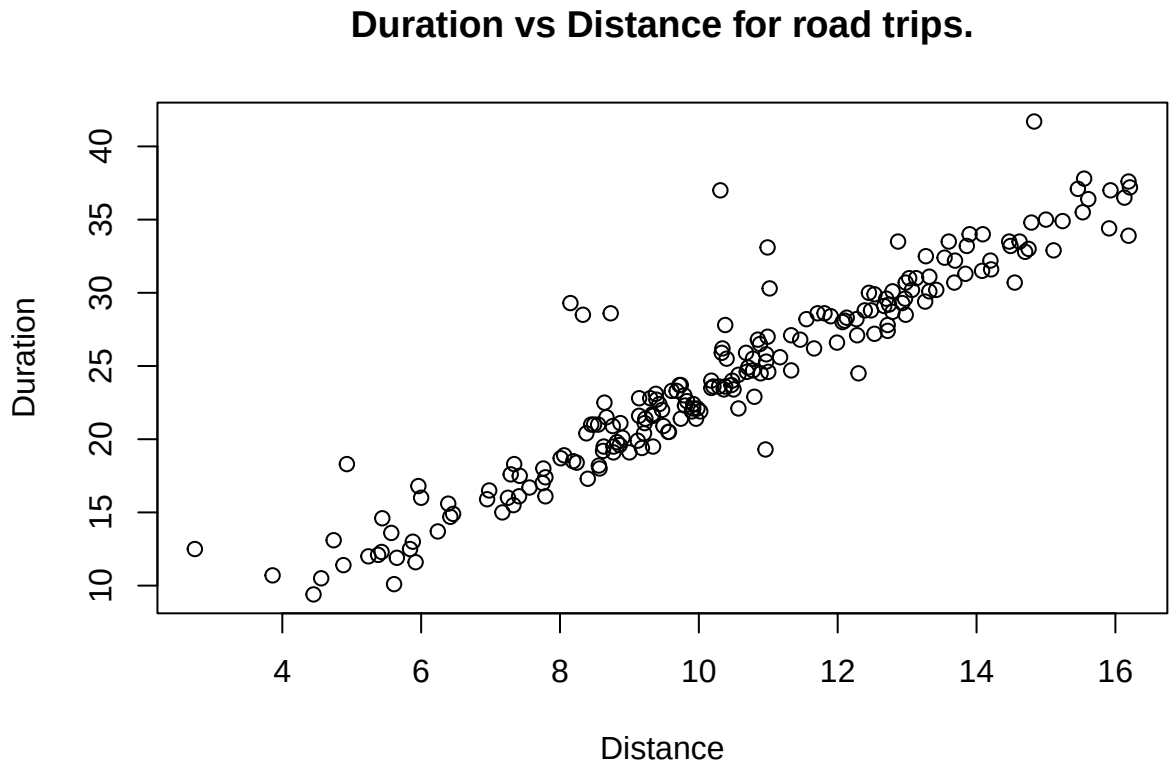
In the case of the duration variable, we see the interval containing 24, but not 23, which explains the different results in b,c contra d. Regarding the distance variable, we see the 95% confidence interval reaching from practically 10.00 to 10.80, so indicating that the μ_s is greater than 10.00, although we get a relatively weak evidence. We also see this since the P-value in this case was not much lower than 0.05.

g. Here we can do

```
with(tripdata, cor(Duration, Distance))
```

```
## [1] 0.9461513
```

```
with(tripdata, plot(Distance, Duration,
                    main = "Duration vs Distance for road trips."))
```



The correlation is about 0.95, and the plot shows strong dependency between the variables.

4.3

a. We read the data in the ordinary way.

```
flats <- read.csv("M:/Undervisning/Undervisningh25/Log708/Data/flat_prices.csv")
head(flats)
```

```
##   price area rooms standard situated town distcen age rent
## 1  1031  100     3        2        6    1       5  15 2051
## 2  1129  116     3        1        5    1       4  42 2834
## 3  1123  110     3        2        5    1       3  25 2468
## 4   607   59     2        3        5    1       6  25 1940
## 5   858   72     2        3        4    1       1  17 1611
## 6   679   64     2        2        3    1       3  17 2039
```

From the (updated) exercise text, we find the encoding for `town` as (1, 2, 3) for (Molde, Kristiansund, Ålesund).

we can use the `tapply` function to compute means in the towns as follows.

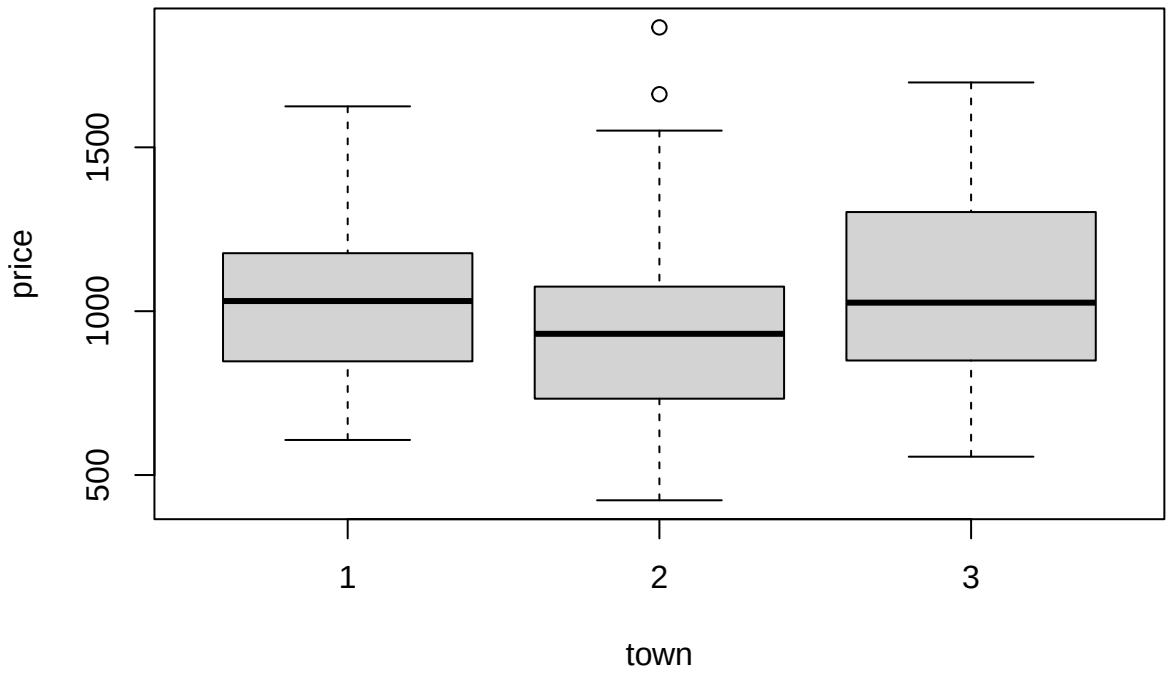
```
with(flats, tapply(price, town, mean))
```

```
##        1        2        3
## 1008.122  949.500 1064.538
```

We see some clear price differences on average.

b. We can do

```
with(flats, boxplot(price ~ town))
```



This points clearly to the same differences, while also showing a slightly more widespread distribution in Ålesund.

c. We want to test whether "true" mean prices are different, i.e. something like

$$H_0 : \mu_M = \mu_K \quad \text{vs} \quad \mu_M \neq \mu_K$$

where the μ's are means in Molde, Kristiansund. We use the `t.test` as suggested, after excluding town 3 from the data. The "two.sided" is the default alternative, and need not be

specified.

```
flats_MK <- subset(flats, town != 3)
with(flats_MK, t.test(price ~  town))
```

```
##
##  Welch Two Sample t-test
##
## data:  price by town
## t = 1.1339, df = 99.663, p-value = 0.2596
## alternative hypothesis: true difference in means between group 1 and group 2 is not e
## 95 percent confidence interval:
##  -43.95448 161.19839
## sample estimates:
## mean in group 1 mean in group 2
##        1008.122         949.500
```

```
# or: t.test(flatsMK$price ~ flatsMK$town)
```

The relatively high P-value means we can not reject the null hypothesis.

For Ålesund, Kristiansund, we run the same procedure

```
flats_KA <- subset(flats, town != 1)
with(flats_KA, t.test(price ~  town))
```

```
##
##  Welch Two Sample t-test
##
## data:  price by town
## t = -1.9751, df = 82.279, p-value = 0.05161
## alternative hypothesis: true difference in means between group 2 and group 3 is not e
## 95 percent confidence interval:
##  -230.8975312    0.8206081
## sample estimates:
## mean in group 2 mean in group 3
##         949.500        1064.538
```

Supposing the significance level is 0.05, the P-value here is at the limit, but still does not lead to a rejected null hypothesis.

d. The variable is called `area`. We can use the method from a).

```
with(flats, tapply(area, town, mean))
```

```
##         1         2         3
##  95.92683 100.47143  98.69231
```

So, on average the sample has somewhat different sized flats.

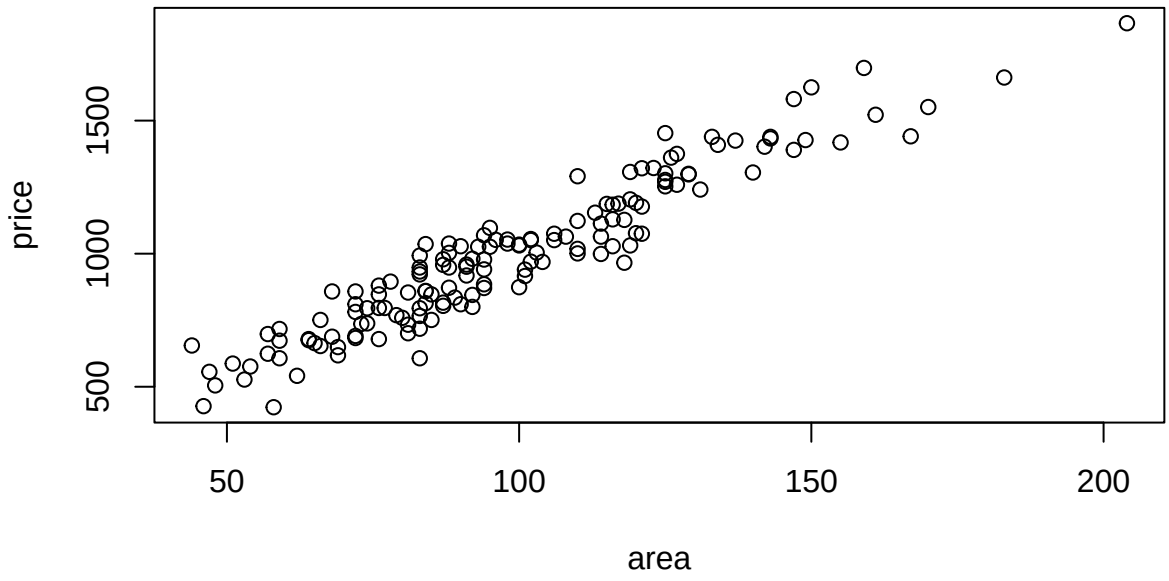
e. For example,

```
with(flats, cor(price, area))
```

```
## [1] 0.9503176
```

Strong correlation, at 0.95. In this connection, showing a scatterplot does not hurt.

```
with(flats, plot(area, price))
```



That looks more or less as expected.

f. We can do the computation as follows. Square meter prices are x1000 NOK

```
flats$sqmprice <- flats$price / flats$area
head(flats)
```

```
##   price area rooms standard situated town distcen age rent  sqmprice
## 1  1031  100     3        2        6    1       5  15 2051 10.310000
## 2  1129  116     3        1        5    1       4  42 2834  9.732759
## 3  1123  110     3        2        5    1       3  25 2468 10.209091
## 4   607   59     2        3        5    1       6  25 1940 10.288136
## 5   858   72     2        3        4    1       1  17 1611 11.916667
## 6   679   64     2        2        3    1       3  17 2039 10.609375
```

g. So, we do a t test again. Note, we need to recalculate the subset to include the `sqmprice` variable. So we can do

```r
flats_MK <- subset(flats, town != 3)
with(flats_MK, t.test(sqmprice ~  town))
```

```
##
##  Welch Two Sample t-test
##
## data:  sqmprice by town
## t = 6.6785, df = 75.642, p-value = 3.581e-09
## alternative hypothesis: true difference in means between group 1 and group 2 is not e
## 95 percent confidence interval:
##  0.7935811 1.4681197
## sample estimates:
## mean in group 1 mean in group 2
##        10.63212         9.50127
```

The P-value is almost 0, and much less than 0.05, so we reject the null hypothesis. We conclude that square meter prices are significantly higher in Molde than in Kristiansund.

h. When comparing nominal prices, we do not take into account that while the prices were on average higher in Molde, the flats were also smaller in Molde. So, comparing nominal prices can be misleading when we don't control for the fact that sizes differ on average. Looking at square meter prices is one way to make a more "fair" comparison as the size is taken into account.

i. So, we can make a few more subsets, and run similar tests:

```r
flats_KA <- subset(flats, town != 1)
flats_MA <- subset(flats, town != 2)

with(flats_KA, t.test(sqmprice ~ town))
```

```
##
##  Welch Two Sample t-test
##
## data:  sqmprice by town
## t = -8.0705, df = 72.688, p-value = 1.066e-11
## alternative hypothesis: true difference in means between group 2 and group 3 is not e
## 95 percent confidence interval:
##  -1.695627 -1.023977
## sample estimates:
## mean in group 2 mean in group 3
##         9.50127        10.86107
```

```r
with(flats_MA, t.test(sqmprice ~ town))
```

```
##
##  Welch Two Sample t-test
```

```
##
## data:  sqmprice by town
## t = -1.1576, df = 77.974, p-value = 0.2505
## alternative hypothesis: true difference in means between group 1 and group 3 is not e
## 95 percent confidence interval:
##  -0.6226931  0.1647894
## sample estimates:
## mean in group 1 mean in group 3
##        10.63212        10.86107
```

We see H_0 rejected when comparing Kristiansund and Ålesund, while not when comparing Molde and Ålesund.

As a final remark, when working with categorical variables like `town` here, it can be worthwile to convert to a "factor". Some code for this is shown in section 7.5 in the compendium. If we do this and rerun question a, we get

```
with(flats, tapply(price, town, mean))
```

```
##    Molde   Krsund  Alesund
## 1008.122  949.500 1064.538
```

So instead of constantly trying to remember what was 1, 2, 3 - we now get the actual names in the output.